Rules, Tables, Graphs, and Sensible Domains

Tables, rules, graphs, solving for the input, and domains that fit the situation

Directions

- A function can be given by a rule, a table, a graph, or a situation. These problems move between all four, so read carefully what each one gives you and what it asks for.
- Filling a table means substituting an *input*; solving means you were handed an *output* and must find the input that produced it.
- Grids are for you to plot on. Label the axes with the quantities they carry.
- When a function models a real situation, ask which inputs actually describe something that could happen.

1. Let $f(x) = 3x - 5$. Complete the table.

x	-2	0	4
$f(x)$	________	________	________

2. Let $g(x) = x^2 - 4$. Find each output. Substitute the input in parentheses first.

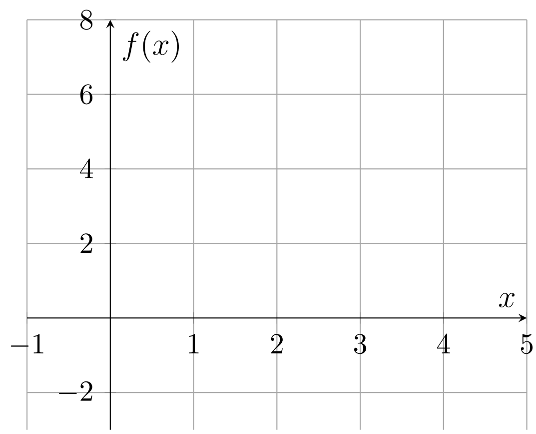
$g(3) =$ ________ $g(-1) =$ ________

3. Let $f(x) = 2x + 9$. Solve $f(x) = 1$.

Answer: ________________

4. Let $f(x) = -2x + 6$. Complete the table, plot the three points on the grid, and draw the straight line through them.

x	0	1	2
$f(x)$	________	________	________



Use your graph: at which input does the line cross the x-axis? $x =$ ________

5. A taxi charges a \$3 pick-up fee plus \$2 for each mile. The fare for a ride of m miles, in dollars, is $C(m) = 3 + 2m$.

(a) Find $C(7)$, the fare for a 7-mile ride, in dollars. $C(7) =$ _________

(b) The rule returns a number when $m = -4$. Explain why -4 is still not in the domain of this model, and state which inputs are.

6. The table shows values of a linear function p.

x	0	2	4
$p(x)$	-3	1	5

(a) Find $p(6)$. $p(6) =$ _________

(b) Find the input whose output is 0. $x =$ _________

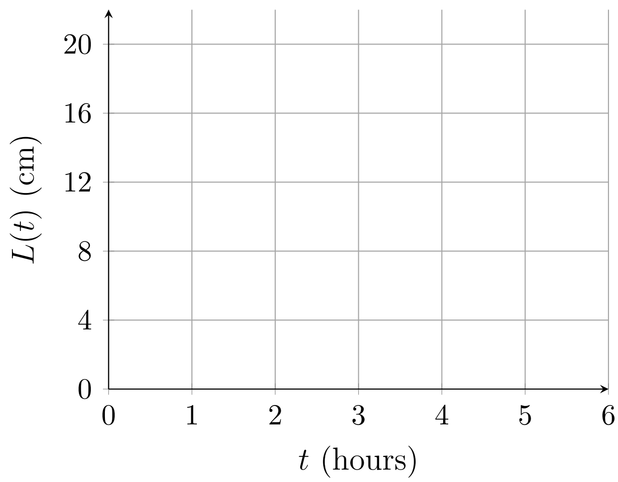
7. Let $f(x) = 8 - 3x$. Solve $f(x) = 23$.

Answer: __________________

8. A candle is 20 cm tall and burns down 4 cm every hour, so its height after t hours is $L(t) = 20 - 4t$ centimetres.

(a) Complete the table, then plot the points and draw the line.

t (hours)	0	2	4
$L(t)$ (cm)	________	________	________



(b) Solve $L(t) = 0$ to find when the candle is used up.

Answer: ________________ hr

9. A school van seats 15 students. A field trip costs \$40 for the driver plus \$6 for each student, so the cost for s students is $C(s) = 40 + 6s$ dollars.

(a) Find $C(9)$, the cost for 9 students, in dollars. $C(9) =$ ________

(b) The rule returns a value for $s = 20.5$. Explain which values of s actually make sense for this trip, and why.

10. The table shows values of a function q.

x	1	2	3	4
$q(x)$	5	8	11	14

(a) Find the constant rate of change of q. rate = ________

(b) Write a rule for $q(x)$, and say how the table shows that q is linear.

(c) Use your rule to find $q(10)$. $q(10) =$ ________

11. Let $f(x) = x^2 - 6x + 5$. Solve $f(x) = 0$. There are two inputs; give both.

Answer: ________________

12. A rectangle has perimeter 40 metres. If one side measures x metres, the other measures $20 - x$ metres, so the area is $A(x) = x(20 - x)$ square metres.

(a) Find $A(6)$, the area when one side is 6 metres.

Answer: ________________ m^2

(b) Explain why the domain of this model is every input strictly between 0 and 20, and not every real number.